Public Vs Private IP addresses

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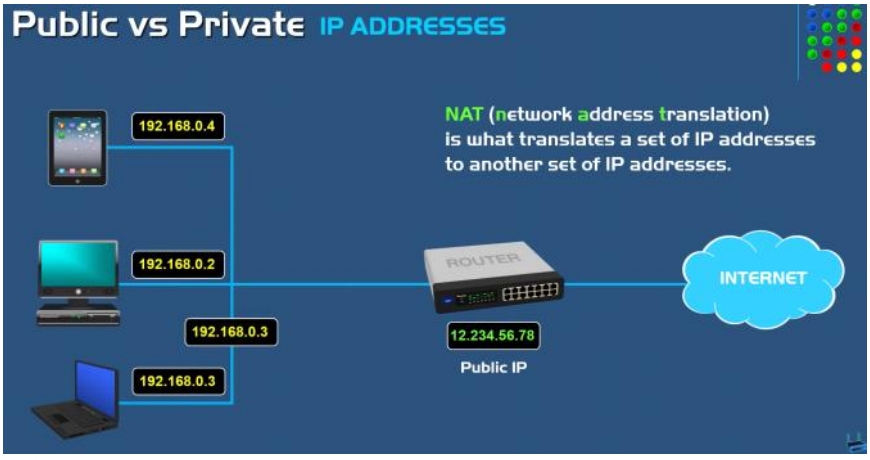
An IP address is a numeric identifier for a computer or device on a network, essential for communication. There are two types: public and private IP addresses.

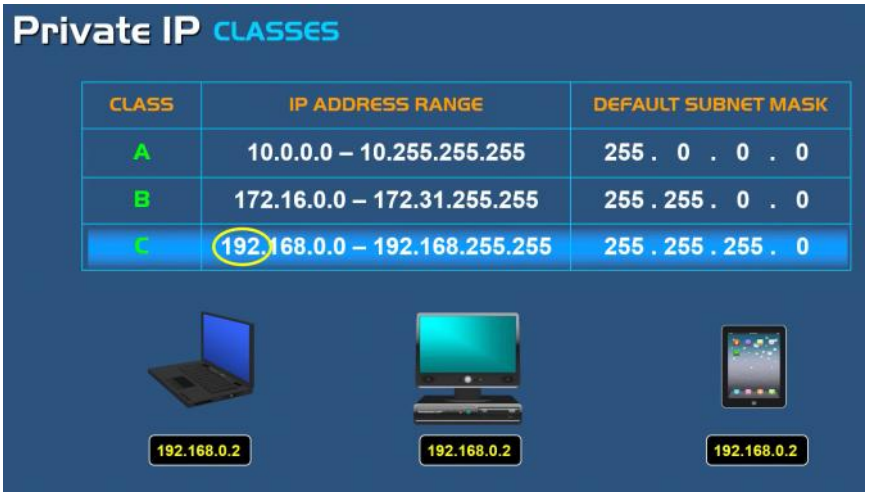
Public IP addresses are assigned by your internet service provider, are registered on the internet, and are what allow you to access the world wide web. They are unique globally.

Private IP addresses are not publicly registered and cannot directly access the internet. They are used internally within a home or business network and are assigned by your router. To access the internet, a private IP address must be converted to a public IP address through a process called Network Address Translation (NAT), which is built into your router. Private IP addresses are non-unique, meaning the same private IP address can be used in different private networks. They are more secure as they are hidden from the internet.

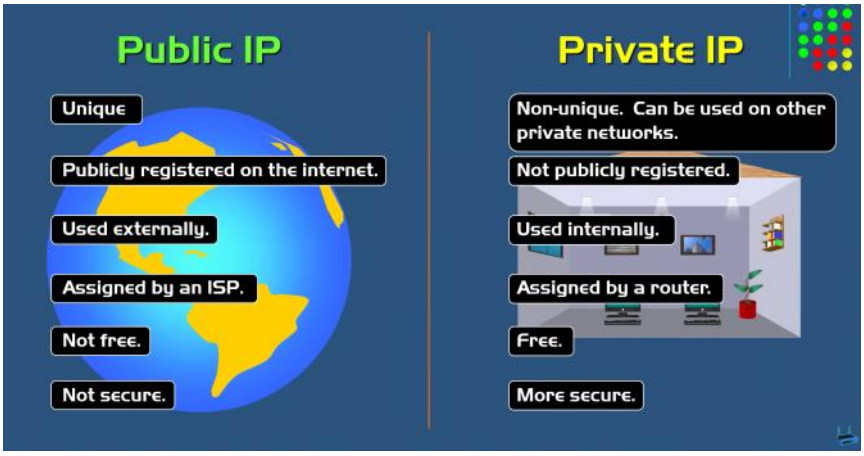


Private IP addresses have three classes:

Class A (starts with 10): Used for large organizations.
Class B (starts with 172): Used for medium-sized organizations.
Class C (starts with 192): Most commonly used for small organizations or homes.



CLASS	IP ADDRESS RANGE	DEFAULT SUBNET MASK
A	10.0.0.0 – 10.255.255.255	255 . 0 . 0 . 0
B	172.16.0.0 – 172.31.255.255	255 . 255 . 0 . 0
C	192.168.0.0 – 192.168.255.255	255 . 255 . 255 . 0



You can find your private IP address on a Windows computer by typing ipconfig in the command prompt, or on Mac and Linux by typing ifconfig in the terminal. To find your public IP address, you can visit a website like whatsmyipaddress.com. For enhanced security and privacy when using a public IP address, a VPN (Virtual Private Network) is recommended.

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Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::e4ed:acbb:4027:c6bb%15
IPv4 Address. . . . . . . . . . . : 192.168.0.241
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Default Gateway . . . . . . . . . : 192.168.0.1
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